National College of Ireland



Research In Computing

Week 2: Research Question

The Scientific Method

"**Research is** the process of gathering information about some subject, analyze them using the scientific method with the intention of increase the stock of knowledge"

--(Wikipedia)



BT Young Scientist and Technology Exhibition

https://www.irishtimes.com/news/science/cork-student-who-designed-software-to-detect-deepfake-videos-wins-2021-btyste-contest-1.4453254

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These include Patrick Collison of Stripe (winner in 2005); Shane Curran, founder of Evervault data privacy company (2017) and Sarah Flannery (1999) who discovered a new way to encrypt data.

Cork student who designed software to detect 'deepfake' videos wins 2021 BTYSTE contest

Greg Tarr is the overall winner of the BT Young Scientist & Technology Exhibition

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Kevin O'Sullivan Environment & Science Editor



Overall winner Gregory Tarr (17), a sixth year student from Bandon Grammar School Co Cork pictured with parents Nita and Richard Tarr. Photograph: Chris Bellew /Fennell Photography

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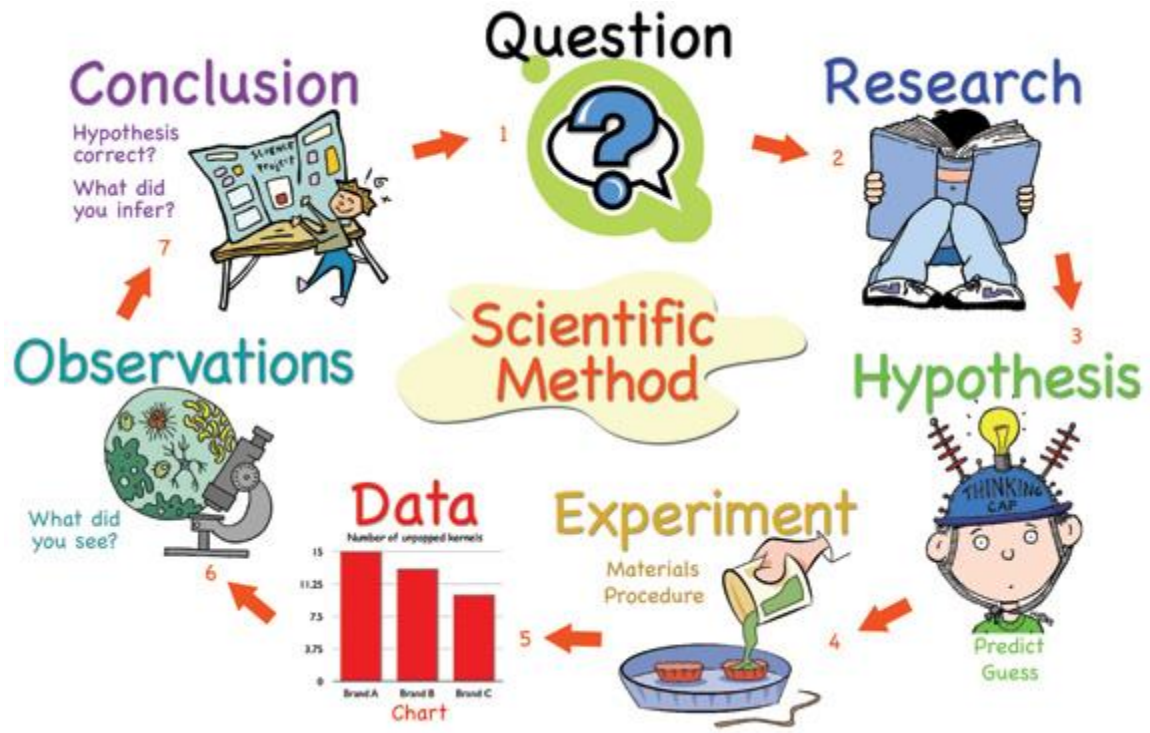
The Scientific Method

"**Research is** the process of gathering information about some subject, analyze them using the scientific method with the intention of increase the stock of knowledge"

--(Wikipedia)

"**The scientific method** is a set of basic rules for a scientist to develop a controlled experiment in order to test and observe events, so that to reach conclusions and report those conclusions, that, in case of validity, are then applied to science"

--(Wikipedia)



The Scientific Method

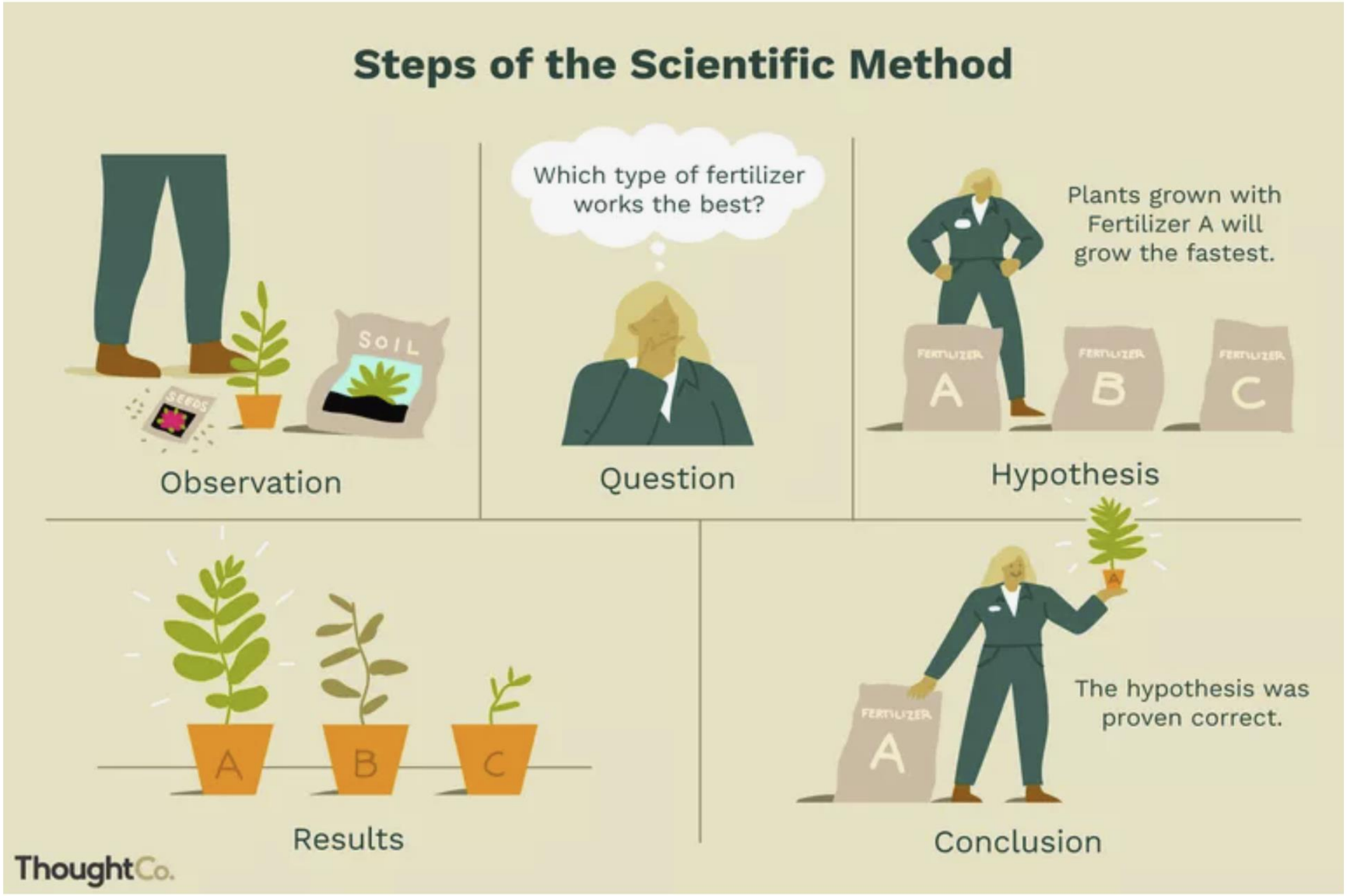
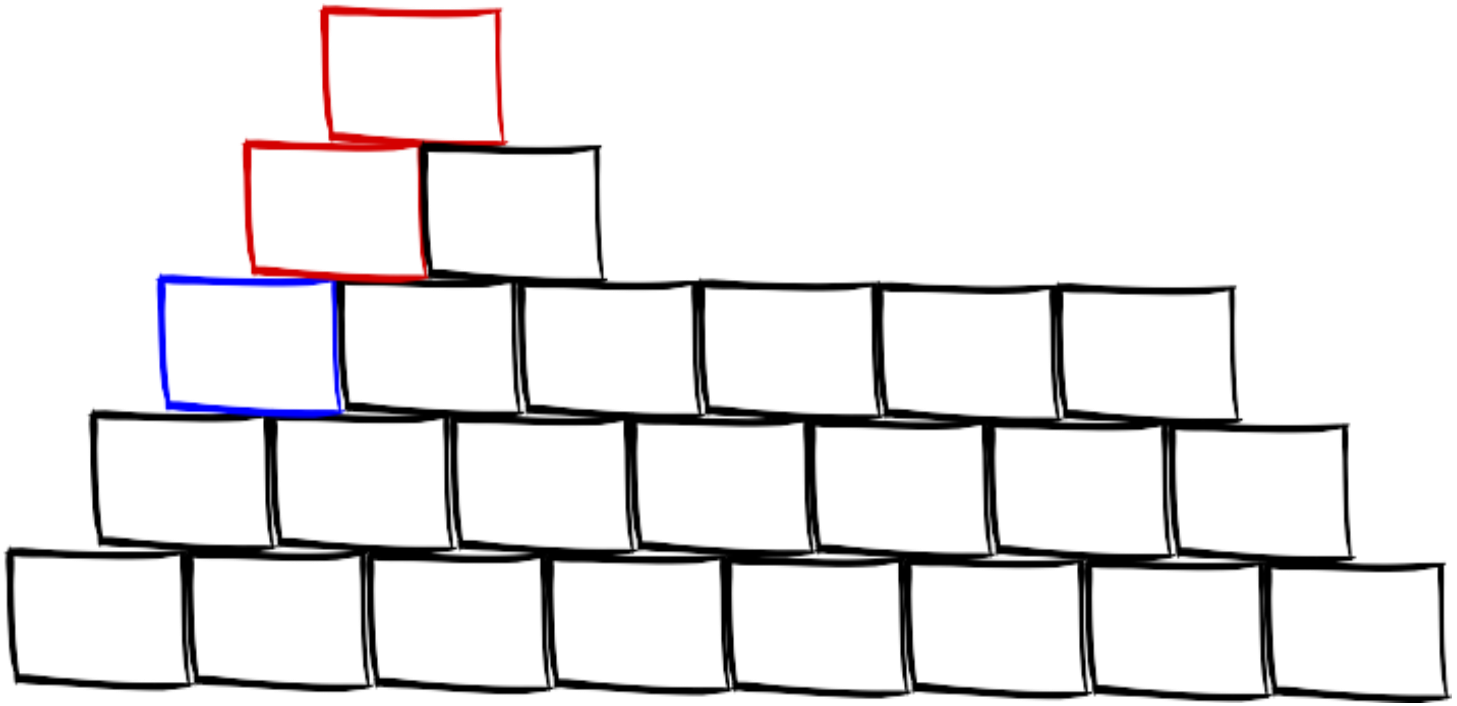


Illustration by J.R. Bee. ThoughtCo.

Research is Incremental





Dimensions in research

- observation vs. innovation
- exploratory vs. exhaustive
- grounded theory vs. case study
- qualitative vs. quantitative
- simulation vs. field trial
- primary sources vs. secondary sources
- theoretical vs. applied
- mathematical vs. empirical
- principled vs. pragmatic

.

What research is not

- `I am planning to build a . . . ‘
- `I am going to find out about . . . ‘
- `I am going to show that A is better than B . . . ‘
- `I think that Q is a good tool for . . . ‘
- `I am going to collect data on . . . ‘
- `I have an interesting algorithm . . . ‘
- `I want to use method X . . . ‘
- `I want to use method X on data Y . . . ‘

From Problem Statement to Research Question

- **Problem Statement**:
  - Identifies a specific issue or challenge in a field.
  - Describes why the issue is significant and needs investigation.
  - Often arises from existing literature, practical challenges, or observed trends.
- **Research Question**:
  - Directly derived from the problem statement.
  - Focuses the investigation to address the core issue.
  - Must be clear, feasible, and significant.
  - Guides the methodology and data collection.

Research Question

- Statement of the goal of a research project
- States clearly what the study will investigate or attempt to prove
- Based on previous discoveries



Purposes of Research Questions

Research question is the backbone of a Master/PhD thesis

Purposes of Research Questions

- They
  - guide your literature review
  - guide your choices in research methodology
  - guide you in deciding what data is needed
  - guide the analysis of your data
  - guide the write-up and interpretation of data
  - stop you following unnecessary directions and tangents



Research is about questions

A good research question:

- Solves a genuine problem, with a genuine challenge. (Avoid work on methods where there is no motivating problem or task, or on solutions to non-problems.)
- Relates to real data, of a realistic volume. Synthetic data allows exploration but is unlikely to confirm predicted behavior on real data.
- Can plausibly be answered in the negative.
- Can plausibly be answered in the scope of your project
- Suggests something useful to measure
- Should have a clear answer
- Has results of interest even if the gains are small

Finding the research question

- Work extensively on a single question during the Research in Computing module
- Make sure the question meets your needs and career prospects
- Match the topic to your strengths – though you may not know what they are yet!
- A single research area can cover many different kinds of topic
- Don't come up with a 'temporary' question just to comply with the requirement of the module

Features of Good Research Questions

- Make a **significant contribution** to knowledge
- **Researchable**: you are able to collect evidence that would answer the question
- **Doable**: given your time and material constraints
- **Inspiring**: potential to hold your interest
- Agood question is situated in the **discourse of science**

Features of Good Research Questions

- **Clear**: it provides enough specifics that one's audience can easily understand its purpose without needing additional explanation
- **Focused**: it is narrow enough that it can be answered thoroughly in the space the writing task allows
- **Concise**: it is expressed in the fewest possible words
- **Complex**: it is not answerable with a simple "yes" or "no," but rather requires synthesis and analysis of ideas and sources prior to composition of an answer
- **Arguable**: its potential answers are open to debate rather than accepted facts

Source: https://writingcenter.gmu.edu/guides/how-to-write-a-research-question

Steps to Developing a RQ

- Choose an interesting general topic
- Do preliminary research on your general topic
- Consider your audience
- Start asking questions
- Evaluate your question
- Begin your research

Source: https://writingcenter.gmu.edu/guides/how-to-write-a-research-question

Defining a RQ (1)

- **Why did the chicken cross the road?**
  - The question is too broad
  - Does not define the segments of the analysis
  - It does not mention which chicken or which road

Defining a RQ (1)

- **How many chickens crossed Stillorgan Road in Dublin, on February 6, 2014?**
    - Simple Answer — perhaps in one sentence
    - No room for analysis

Defining a RQ (1)

- **What are some of the environmental factors that occurred in Dublin between January and February 2014 that would cause chickens to cross Stillorgan Road?**
    - More precise
    - Author can argue which factors are significant
    - Author can argue how results are beneficial (or not)
    - Leaves space for analysis

Defining a RQ (2)

- **What is the effect of global warming on the environment?**
  - Too broad! You can write many books about this.
  - Find focus, e.g., by selecting a **relevant** species

Defining a RQ (2)

- **How is glacial melting affecting penguins?**
    - Still to broad? Pick a **relevant** location

Defining a RQ (2)

- **How is glacial melting affecting penguins in the Antarctica?**
  - Location and factor specific influencing penguins in general
  - There could be another refinement specifying exactly the aspect of penguins' life of interest
  - However, this level of detail gives some flexibility to explore how glacial melting affects different aspects of penguins' life

Defining a RQ (3)

- **Why are social networking sites harmful?**
  - “Harm” is subjective. Are you preoccupied? If so, change the topic!
  - “Harm” is undefined. What does it mean? Harm for whom?
  - “Social media sites” is too broad. Narrow it down. Find one aspect

Defining a RQ (3)

- **How do people experience privacy issues on social networking sites?**
  - Better, but what **people**? Maybe be a bit more specific here.

Defining a RQ (3)

- **How are online users experiencing privacy issues on social networking?**
  - Do you want to narrow down more, e.g., pick a particular social media site?
  - You can!

Defining a RQ (3)

- **How are online users experiencing privacy issues on social networking sites like Facebook?**
    - Do you want to bring in another aspect to widen your topic a little? Sure!

Defining a RQ (3)

- **How are online users experiencing or addressing privacy issues on social networking sites like Facebook?**

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Example: web search

- **Statistical:** Identify properties of web pages that are useful in determining whether they are good answers
- **Mathematical:** Prove that the efficiency of index construction has reached a lower bound in terms of asymptotic cost
- **Analytical:** Quantify bottlenecks in query processing, relate them to machine and network properties
- **Algorithmic:** Develop and demonstrate the benefit of a new index structure
- **Representational:** Develop a formal language for capturing properties of image, video, audio, etc., to be used in search
- **Behavioral:** Quantify the effect of varying the search interface
- **Social:** Link changes in search technology to changes in public opinion on key topics

Example: web search

- Notice something: creating a search engine that works on the same principles as current search engines is **not a research question**
- Notice also that creating a (good) search engine is a **difficult task**, but it is not a research question
- A research question needs to answer an **intellectual challenge**

If there are books/articles on how to do it then it is not a research question. Even if it is extremely difficult to do!

Another example: creating an operating system.

Framing the question

Expect development of a precise statement of the question to take time.

- It is common to have to revisit the question after some substantial work has been done
- Development of a question is a good basis for early supervision meetings
- Consider alternatives, and their strengths and shortcomings
- This may be how you discover whether your supervisor makes sense for you as a (senior) research partner

Framing the question

Consider whether:

- Answering the question involves access to data and if so, whether the necessary data can be obtained or created
- The volume and diversity of the research literature is likely to be manageable
- There is a feasible methodology for pursuing an investigation